

Enhancing the Immersive Experience of the *Yijing* in Claborate-Style Painting through Virtual Reality

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ABSTRACT

Chinese claborate-style painting is a traditional art form that depicts objects with delicate brushstrokes of color. The exhibition of Chinese paintings can be enhanced by the application of virtual reality (VR) by creating an immersive virtual environment and engaging experience for viewers. In order to convey the *yijing* (artistic conception in Chinese painting) of the claborate-style painting in VR, this paper investigates how to enhance the viewer's feeling of *yijing* in VR and sets up two virtual viewing paths to provide the viewer with a more immersive experience. This study can contribute to the emerging field by offering new insights into *yijing* in VR and providing a new reference for future VR design practices of Chinese paintings.

Keywords: Virtual reality, claborate-style painting, 2.5D, Chinese painting, *yijing*, artistic conception.

Index Terms: Applied computing—Arts and humanities—Media arts; Human-computer interaction (HCI)—HCI design and evaluation methods—User studies

1. INTRODUCTION

Chinese claborate-style painting is a type of Chinese painting that depicts objects through delicate brushwork. Due to the delicate nature of these paintings, they are often safeguarded behind glass cases in museums to ensure their preservation. However, this can make it difficult for viewers to see the details of the painting up close. The development of science and technology has opened up new possibilities for artistic expression. For example, the virtual reality (VR) representation of the Chinese painting 'Spring Morning in the Han Palace' was used to learn about Chinese art and acquire cultural and historical insights into the understanding of Chinese painting. The experiment demonstrated that the VR environment significantly increased learning effectiveness and motivation [1].

However, to better integrate art and technology, it is crucial to accurately recreate the *yijing* (artistic conception) of Chinese paintings in the virtual world through VR. *Yijing* refers to a poetic realm that consists of a series of images where *yi* and *jing* merge. *Yijing* can be translated as "the realm of meaning". *Yi* denotes "meaning" or "idea", and *jing* means "realm" or "domain" [2]. Zong Baihua is a pioneer in the study of ancient Chinese aesthetics, and he referred to the *yijing* of Chinese painters and poets as 'the center of artistic creation' [3].

Previous research on combining Chinese painting and VR involves developing a VR-based exhibition system to showcase traditional Chinese Hand-Scroll Painting. The systems aim to recreate the real-world viewing experience of the ancient and the narrative-world touring experience [4]. Another research focuses on digitizing traditional paintings and transitioning from 2D to 3D content immersive artworks in VR [5]. It is still being determined

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whether VR can restore the artistic conception of Chinese paintings, as there is no relevant research. Therefore, investigating how VR affects the user's perception of *yijing* in paintings is meaningful.

This research explores users' perception of the *yijing* of Chinese claborate-style paintings using VR technology. The painting selected in this study is 'Bamboo Courtyard Pin Gu painting,' a screen-type landscape painting that constructs a three-dimensional space within a two-dimensional plane [6], from Qiu Ying, a famous painter of claborate-style painting in the Ming dynasty. This paper presents a 2.5D immersive representation of this painting in VR. Two types of viewing paths, painter-path and free-path, are designed for the audience to view the painting in a virtual environment. A user study was carried out by integrating the VR painting with the traditional paper painting. One main contribution of this study is that it offers novel insights into viewers' feelings of *yijing* in VR painting. Moreover, it proposes the 2.5D immersive representation of Chinese claborate-style painting that can serve as a new reference for future VR design practices in this field.

2. METHOD

The user study of this research was designed to explore whether VR can help people better perceive the *yijing* of the paintings after viewing the paper painting.

2.1 IMMERSIVE 2.5D DISPLAY

This study creates the immersive display of the claborate-style painting based on the concept of 2.5D, as shown in Figure 1. 2.5D refers to a 3D environment created using 2D projection, which involves arranging the painting's elements in layers to produce a visual illusion like a stage. The production process includes: 1) The painting is divided into different elements; 2) The obscured part of the element needs to be complemented to ensure that each element is complete and independent; 3) Photoshop was used to create the 2.5D model of elements; 4) All 2.5D models were imported into Unity, and the distance between each 2.5D model, spatial depth, and relative position relationship in Unity were adjusted to restore the original painting arrangement to the maximum extent.



Figure 1: 2.5D immersive display of the claborate-style painting

2.2 VIEWING PATHS

Two viewing paths are designed for experiencing the VR painting: a painter-path and a free-path. Inspired by the Chinese concept of "one step makes a difference", the painter-path provides a predetermined route for users to view the painting as an expert (painter). This path is designed based on Zong Baihua's expression of *yijing* and incorporates his feelings about art to evaluate the

users' feelings about the *yijing*. Users can navigate through the immersive environment following the predetermined spots. In the free-path mode, there is no pre-defined route for users. They can choose their viewing path and explore the painting at their own pace by navigating through the immersive environment.

2.3 EXPERIMENT

Twenty subjects (10 males and 10 females) were recruited for the experiment. They were tasked with two activities: viewing the print copy of the painting and viewing the same painting in VR. When they finished each task, they had to fill in a questionnaire.

In the first task, the subjects viewed a high-resolution print of the painting (Figure 2, left) with unlimited but recorded time for the subjects to enjoy the painting. Then, they rated their perception of *yijing* through a questionnaire. In the second task, the subjects received brief instructions on the operation in VR and wore the VR headset (Oculus Quest 2) to appreciate the painting through the two viewing paths provided (painter-path and free-path) (Figure 2, right), with unlimited but recorded viewing time. Subjects completed a questionnaire after the second task covering *yijing* perception, VR experience (including engagement, presence, immersion, and emotion), and the System Usability Scale (SUS) for the usability test. The questionnaires used a Likert scale ranging from 'strongly disagree' (score 1) to 'strongly agree' (score 5).



Figure 2: Viewing the paper painting (left) and viewing the VR painting (right)

The questionnaire assessing user perception of *yijing* towards the painting was designed with three aspects, and the scores of the questions in each aspect were counted as an average. The three aspects were virtuality-reality combination (A1), rhythmic vitality (A2), and spatial view in Chinese painting (A3). Each question in the *yijing* questionnaire is derived from Zong Baihua's expression of insights into *yijing* and his sentiments about art [3]. The questions are structured to guide respondents through a three-step progression from observation to understanding to final revelation.

3. RESULTS AND DISCUSSION

Through experiments, as shown in Figure 3, it is found that integrating the VR painting with the traditional paper painting can significantly improve the viewers' feeling of *yijing* towards the painting. *Yijing's* rating in the virtuality-reality combination (A1) increased from $M=2.98$ to 4.05. This is an expected enhancement because VR can establish an extremely strong virtuality through the immersive virtual environment. It is interesting to find that there are also improvements in the other two aspects of *yijing*. The rating of rhythmic vitality (A2) rises from $M=3.30$ to 4.07, and the spatial view in Chinese painting (A3) is enhanced from $M=3.33$ to 4.38. According to One-Way ANOVA, significant differences in rating results were found in the three aspects ($p < 0.005$). Therefore, according to the experimental finding, viewing the elaborate-style painting in VR helps the subjects better perceive the *yijing* in painting in all three aspects.

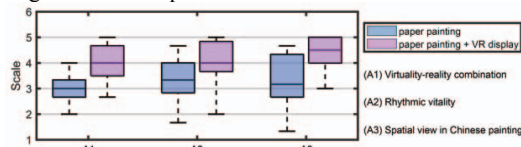


Figure 3: The boxplot of the paper painting and VR painting

In the VR experience, the subjects rate engagement and presence as the highest scores (both $M=4.50$), followed by immersion ($M=4.25$), which indicates the subjects enjoy the sense of involvement, emphasize the feelings of being there and immerse themselves in VR painting. The lowest rating is found in emotion ($M=4.07$), indicating that the current VR painting display can be further enhanced in evoking emotional responses from participants.

When the subjects viewed the traditional paper painting, the average viewing duration for the 20 subjects was around 15 seconds. In contrast, during the VR experience, the average time extended to about 10 minutes, significantly prolonging the subjects' viewing time. During interviews, subjects expressed that they had completely forgotten about time and had just immersed themselves in the VR experience. In the final interview at the end of all the experiments, subjects noted that the painter-path offered by the VR application helped them to sort through the painting quickly and more methodically, which gave them a basis for understanding the painting. Meanwhile, the free-path provided greater autonomy for subjects. The two types of paths, painter-path and free-path, complement each other effectively and create different and engaging viewing experiences for users.

Considering the usability of the VR painting display, a System Usability Scale (SUS) analysis was conducted, yielding a mean score of 70.5, indicating that the VR painting display proposed in this study reaches the average and the method is feasible. However, it is noteworthy that SUS scores fluctuate widely among subjects (ranging from 50 to 85). Therefore, introducing VR painting in the future necessitates enhanced guidance and instructions for users.

4. CONCLUSION

In this paper, the authors mainly investigated the immersive display of the Chinese elaborate-style painting and the viewer's perception of *yijing* in VR painting. Two types of viewing paths were established in the VR environment: painter-path and free-path. The viewer's feeling of *yijing* was evaluated through three key aspects: virtuality-reality combination, rhythmic vitality, and spatial view in Chinese painting. The results indicate that VR technology can improve subjects' perception of the three aspects of *yijing* in elaborate-style painting compared to viewing the paper painting alone. The subjects expressed a positive attitude towards VR through questionnaires and interviews, suggesting that VR can increase their willingness to enjoy the elaborate-style painting and deepen their understanding of *yijing*.

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